

World Wide Web



What is the Internet?

The Internet is the “network of networks”

It was born in the USA in 1969, created by the *Department of Defense Advanced Research Projects Agency NET* (DARPANET)

Especially famous for the Web and E-mail services, but the Internet is (or was) also:

- FTP (File Transfer Protocol)
- Telnet (Terminal emulation)
- Newsgroups
- VOIP
- ...

Important: Internet ≠ Web!

A short history of the Internet...

1961: J.C.R. Licklider (M.I.T.) introduces the concept of *network* (interconnected computers)

1966: the foundations of the ARPANET network are laid (developed by the Advanced Research Projects Agency = ARPA, subsequently DARPA = Defence Advanced...)

1972: Robert Khan organizes the first public demonstration of the ARPANET at an international conference

1973: Robert Khan and Vinton Cerf start the definition of TCP/IP (Transmission Control Protocol/Internet Protocol)

1980: TCP/IP is adopted by the DARPANET

... a short history of the Internet...

1983: several communities already use the DARPANET, which, however, is not the only existing network (examples of other networks, all “purpose-built”, are: MFENet of the Energy Department, SPAN of NASA, CSNET of the academic and industrial communities, USENET of AT&T, ...)

1985: the NSFNET network is specifically created to connect universities each other; the TCP/IP protocol is adopted

End of 80s: the Internet (“network of networks”) takes shape

1990: the foundations for the World Wide Web are laid

... a short history of the Internet

1990: Tim Berners-Lee and Robert Cailliau (CERN, European Laboratory for Particle Physics, in Geneva, Switzerland) present the proposal for a communication system based on *hypertexts*

1991: the Stanford Linear Accelerator Center in California (SLAC) is the first web server in the USA

1992: CERN releases a free web browser; several research centers and universities in the world “become connected”

1993: in October, there are about 200 web servers worldwide

1994: *Mosaic Communications Corporation* is born (which, subsequently, will become *Netscape*)

After **1995:** the “browsers war” begins (*Netscape Navigator* vs. *MS Explorer*)

The World Wide Web

The Web is an Internet-based system that exploits hypertext technology to distribute documents, files, images, etc.

The Web is a **service** of the Internet

It was born in **1989** at **CERN**, in **Geneva** (Switzerland), as a way to quickly **share scientific documents**



Tim Berners-Lee

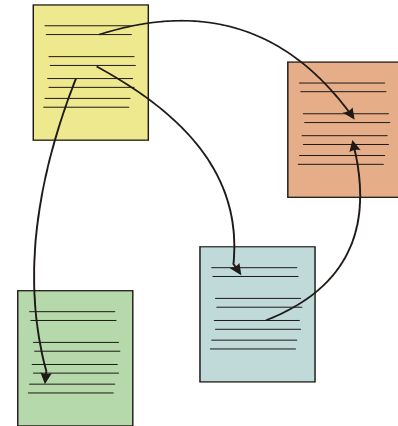
Hypertext and Hypermedia

Hypertext

Text that contains *links* to other documents – It provides a method for non-sequential browsing

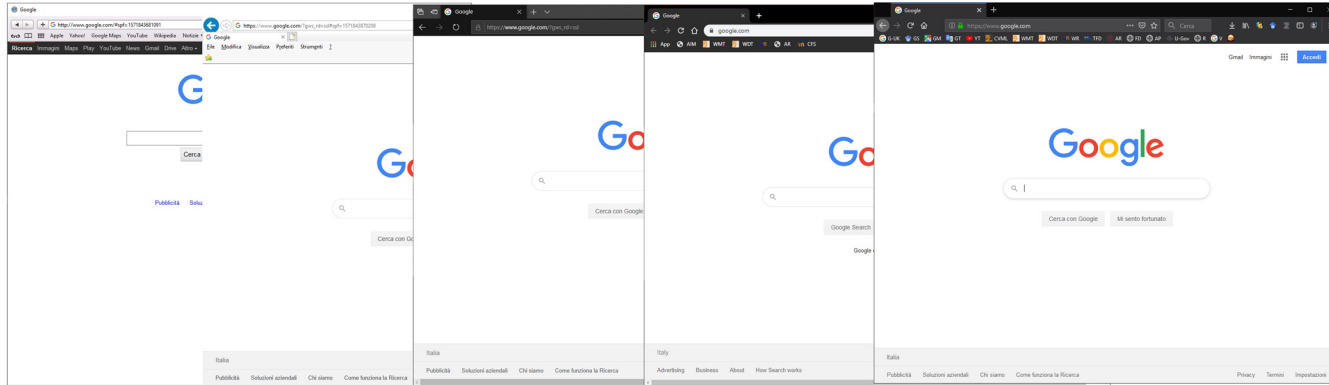
Hypermedia

Hypertext with links to multimedia elements (images, videos, etc.)



The web browser

Is the tool (software) that enables web surfing
do all browsers behave the same way?



Unfortunately, the answer is (still now) no... not exactly!

URL

URL = Uniform Resource Locator

Is the (unique) address of a file available on the Internet

Example:

http://www.ncsa.uiuc.edu/General/Internet/WWW/HTMLPrimerAll.html

The diagram shows the URL 'http://www.ncsa.uiuc.edu/General/Internet/WWW/HTMLPrimerAll.html' with four labels below it, each with an upward-pointing arrow: 'protocol' points to 'http://', 'server address' points to 'www.ncsa.uiuc.edu', 'path' points to '/General/Internet/WWW/' (indicated by a bracket above the path), and 'document' points to 'HTMLPrimerAll.html'.

HTTP = Hyper Text Transfer Protocol

HTTPS = secure version of HTTP (the current standard)

Client-Server architecture...

Server

- Hardware that provides some kind of service (e.g., HTTP, FTP, ...)
- Software that manages some kind of service
Uses specific ports (e.g., HTTP → 80, Telnet → 23, ...)

Client

- Hardware that uses some kind of remote service
E.g., user's PC
- Software that manages data coming from and sent to some kind of remote service
E.g., web browser

... Client-Server architecture

The Web is based on a typical client-server architecture

Clients (users' computers) request services (web pages) to servers (web servers)

